

Day 4 Problem Set

Geometry Summer Credit | Congruence and Triangle Angles

Name:	Date:	Period:
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Goal

Practice identifying rigid motions, congruent corresponding parts, triangle classification, and angle relationships.

Support Practice

1.	List the three rigid motions.
2.	Is a dilation by scale factor 4 a rigid motion? Explain briefly.
3.	Find the missing triangle angle if two angles are 55 degrees and 75 degrees.
4.	Solve $2x + 30 = 180$.
5.	An exterior angle is 110 degrees. One remote interior angle is 45 degrees. Find the other.

Core Practice: Congruence and Rigid Motions

6.	A triangle is translated 8 units left and 3 units up. Are the original and image congruent? Explain.
7.	A quadrilateral is reflected across the y-axis. What measurements are preserved?

8. Triangle ABC is congruent to triangle JKL in that order. If $AB = 12$, $BC = 15$, and angle $C = 38$ degrees, name the corresponding side and angle measures in triangle JKL.

9. Triangle DEF is rotated 180 degrees to $D'E'F'$. If angle $E = 72$ degrees and $EF = 9$, what are angle E' and $E'F'$?

10. Error analysis: A student says two figures are congruent if they have the same shape, even if one is twice as large. Explain and correct the statement.

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Continued

Core Practice: Triangle Classification and Angles

11. Classify by sides: 5, 5, 8.

12. Classify by sides: 4, 6, 9.

13. Classify by angles: 25 degrees, 65 degrees, 90 degrees.

14. Classify by angles: 48 degrees, 58 degrees, 74 degrees.

15. Find the missing angle: 27 degrees, 116 degrees, and x .

16. Triangle angles are $x + 10$, $2x$, and $3x - 4$. Find x and all three angle measures.

17. An exterior angle is 140 degrees. One remote interior angle is 63 degrees. Find the other remote interior angle.

18. Two remote interior angles are 48 degrees and 76 degrees. Find the exterior angle.

19. Error analysis: A student says the exterior angle and one remote interior angle add to 180 degrees. Explain the mistake.

Mixed Review and Reasoning

20. Spiral: Rotate $(-5, 2)$ 90 degrees counterclockwise about the origin.

21. Spiral: Reflect $(6, -4)$ across the x-axis.

22. Challenge: Can two triangles be similar but not congruent? Explain using transformations.

Before You Submit

Circle one congruence problem and one triangle-angle problem you can explain to a partner.